Probleme pentru pregătirea examenului final la Analiză Matematică

1. Să se calculeze următoarele integrale improprii:

a) $\displaystyle\int_0^\infty \frac{\mathrm{d}x}{1+x^4}$; b) $\displaystyle\int_0^\infty \frac{\mathrm{d}x}{\sqrt[3]{x}(1+x^2)}$; c) $\displaystyle\int_0^\infty \frac{\mathrm{d}x}{\sqrt[3]{x^2}(1+x^3)}$;

d) $\displaystyle\int_0^\infty \frac{\sqrt[4]{x}}{(1+x)^2}\,\mathrm{d}x$; e) $\displaystyle\int_0^\infty x^p e^{-x^q}\,\mathrm{d}x,\ p>-1, q>0$;

f) $\displaystyle\int_0^\infty e^{-x^2}\,\mathrm{d}x$; g) $\displaystyle\int_0^{\frac{\pi}{2}} \sin^6 x\cos^4 x\,\mathrm{d}x$; h) $\displaystyle\int_0^1 \left(\ln\frac{1}{x}\right)^p \mathrm{d}x,\ p>-1$.

2. Să se calculeze următoarele integrale folosind derivarea sub semnul integral:

a) $\displaystyle\int_0^{\frac{\pi}{2}} \ln\left(\cos^2 x + y^2\sin^2 x\right)\mathrm{d}x$, cu $y>0$;

b) $\displaystyle\int_0^{\frac{\pi}{2}} \frac{1}{\sin x}\ln\frac{1+y\sin x}{1-y\sin x}\,\mathrm{d}x$, cu $|y|<1$;

c) $\displaystyle\int_0^{\frac{\pi}{2}} \frac{\operatorname{arctg}(y\operatorname{tg}x)}{\operatorname{tg}x}\,\mathrm{d}x$;

d) $\displaystyle\int_0^1 \frac{\ln(1-y^2x^2)}{x^2\sqrt{1-x^2}}\,\mathrm{d}x$, cu $|y|<1$.

3. Să se calculeze lungimile următoarelor curbe:

a) $C:\begin{cases} x=a(t-\sin t),\\ y=a(1-\cos t),\end{cases}\quad t\in[0,2\pi]$, cu $a>0$;

b) $C:\begin{cases} x=a(2\cos t-\cos 2t),\\ y=a(2\sin t-\sin 2t),\end{cases}\quad t\in[0,2\pi]$, cu $a>0$;

c) $C:\begin{cases} x=ae^{mt}\cos t,\\ y=ae^{mt}\sin t,\end{cases}\quad t\in[0,2\pi]$, cu $a,m>0$.

4. Să se calculeze următoarele integrale curbilinii de prima speţă:

a) $\displaystyle\int_C ye^{-x}\,\mathrm{d}s$, unde $C:\begin{cases} x=\ln\left(1+t^2\right),\\ y=2\operatorname{arctg}t-t,\end{cases}\quad t\in[0,1]$;

b) $\displaystyle\int_C \left(x^2+y^2\right)\mathrm{d}s$, unde C este segmentul $[AB]$, cu $A(a,a)$ şi $B(b,b)$, $b>a$;

c) $\displaystyle\int_C xy\,\mathrm{d}s$, unde $C:\begin{cases} x=a\cos^2 t,\\ y=a\cos t,\end{cases}\quad t\in[0,\pi]$, cu $a>0$;

d) $\displaystyle\int_C x\sqrt{x^2-y^2}\,\mathrm{d}s$, unde C este definită de $(x^2+y^2)^2=a^2(x^2-y^2)$, $x\geq 0$, cu $a>0$;

e) $\displaystyle\int_C \left(2z-\sqrt{x^2+y^2}\right)\mathrm{d}s$, unde $C:\begin{cases} x=t\cos t,\\ y=t\sin t,\\ z=t,\end{cases}\quad t\in[0,2\pi]$;

f) $\displaystyle\int_C \frac{\mathrm{d}s}{(x^2+y^2+z^2)^3}$, unde C este definită de $x^2+y^2+z^2=2a^2$ şi $z=a>0$;

g) $\displaystyle\int_C ze^{\sqrt{x^2+y^2}}\,\mathrm{d}s$, unde C este definită de $x^2+y^2+z^2=a^2$ şi $x+y=0$, cu $a>0$.

$$\int_a^b P(x(t),y(t))\cdot\sqrt{(x'(t))^2+(y'(t))^2}\,dt$$

Integrala curbilinie de prima speță

5. Să se calculeze următoarele integrale curbilinii de speţa a doua:

a) $\displaystyle\int_C (2a-y)\,\mathrm{d}x + x\,\mathrm{d}y$, unde $C: \begin{cases} x = a(t-\sin t), \\ y = a(1-\cos t), \end{cases} \quad t \in [0, 2\pi]$;

b) $\displaystyle\int_C \frac{x^3\,\mathrm{d}y - y^3\,\mathrm{d}x}{x^{\frac{5}{3}} + y^{\frac{5}{3}}}$, unde C este arcul $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$, cuprins în primul cadran;

c) $\displaystyle\int_C (y^2+z^2)\,\mathrm{d}x + (z^2+x^2)\,\mathrm{d}y + (x^2+y^2)\,\mathrm{d}z$, unde C este curba de intersecţie a suprafeţelor $x^2+y^2+z^2 = 4ax$ şi $x^2+y^2 = ax$, cu $z \geq 0$;

d) $\displaystyle\int_C y^2\,\mathrm{d}x + z^2\,\mathrm{d}y + x^2\,\mathrm{d}z$, unde C este curba de intersecţie a suprafeţelor $x^2+y^2+z^2 = a^2$ şi $x^2+y^2 = ax$, cu $z \geq 0$;

e) $\displaystyle\int_C (e^x\cos y + xy^2)\,\mathrm{d}x - (e^x \sin y + x^2 y)\,\mathrm{d}y$, unde C este frontiera compactului definit de $x \geq 0, y \geq 0, r \leq a\sqrt{\cos 2t}$, cu $a > 0$;

f) $\displaystyle\int_C (xy+x+y)\,\mathrm{d}x + (xy+x-y)\,\mathrm{d}y$, unde C este curba definită de $x^2+y^2 = 4x$;

g) $\displaystyle\int_C \frac{\mathrm{d}x + \mathrm{d}y}{\sqrt{x^2+y^2+z^2}} + xyz\,\mathrm{d}z$, unde $C: \begin{cases} x = ae^{-t}\cos t, \\ y = ae^{-t}\sin t, \\ z = be^{-t}, \end{cases} \quad t \in [0, 2\pi]$, cu $a, b > 0$;

h) $\displaystyle\int_C y\,\mathrm{d}x + z^2\,\mathrm{d}y + x\,\mathrm{d}z$, unde C este curba de intersecţie a suprafeţelor $z = \sqrt{x^2+y^2}$ şi $z = 6 - (x^2+y^2)$.

6. Să se arate că $\displaystyle\int_C \left(e^{-x^2+y^2}\cos 2xy\right)\,\mathrm{d}x + \left(e^{-x^2+y^2}\sin 2xy\right)\,\mathrm{d}y = 0$, pentru orice curbă netedă, simplă şi închisă, C.

7. Să se calculeze $\displaystyle\int_C y\,\mathrm{d}x + x\,\mathrm{d}y$ pe un drum cu capetele $A(2,1)$ şi $B(1,3)$.

8. Să se calculeze $\displaystyle\int_C \frac{y\,\mathrm{d}x + x\,\mathrm{d}y}{1+xy}$ pe un drum arbitrar cu capetele $A(-1,-1)$ şi $B(3,3)$, ce nu intersectează hiperbola de ecuaţie $xy = -1$.

9. Să se calculeze circulaţia câmpului $\overline{V}(x,y) = \frac{3x^3+xy^2}{\sqrt{x^2+y^2}}\,\overline{i} - \frac{3y^3+x^2y}{\sqrt{x^2+y^2}}\,\overline{j}$ de-a lungul curbei $r = a(1+\cos t)$, cu $t \in \left[0, \frac{\pi}{2}\right]$.

10. Să se calculeze următoarele integrale duble:

a) $\displaystyle\iint_D e^{x^2+y^2}\,\mathrm{d}x\,\mathrm{d}y$, unde $D = \left\{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \leq 1\right\}$;

b) $\displaystyle\iint_D \sqrt{\frac{1-x^2-y^2}{1+x^2+y^2}}\,\mathrm{d}x\,\mathrm{d}y$, unde $D = \left\{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \leq 1, x, y \geq 0\right\}$;

c) $\displaystyle\iint_D \frac{xy\,\mathrm{d}x\,\mathrm{d}y}{\sqrt{x^2+y^2}}$, unde $D = \left\{(x,y) \in \mathbb{R}^2 \mid \frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1, x, y \geq 0\right\}$;

d) $\displaystyle\iint_D e^{-\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}\right)}\,\mathrm{d}x\,\mathrm{d}y$, unde D este exteriorul elipsei $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$;

e) $\displaystyle\iint_D \frac{\mathrm{d}x\,\mathrm{d}y}{\sqrt{x^2+y^2}(1+x^2+y^2)^2}$, unde D este interiorul cercului $x^2+y^2 = a^2$;

f) $\iint_D xy\sqrt{x^2+4y^2}\,\mathrm{d}x\,\mathrm{d}y$, unde D este sfertul din primul cadran al discului unitar;

g) $\iint_D \left(1+\sqrt{x^2+y^2}\right)\mathrm{d}x\,\mathrm{d}y$, unde $D=\left\{(x,y)\in\mathbb{R}^2 \mid x^2+y^2\le y, x\ge 0\right\}$;

h) $\iint_D \dfrac{x^2+y^2}{x+\sqrt{x^2+y^2}}\,\mathrm{d}x\,\mathrm{d}y$, unde D este definit de $y\ge 0$ şi $0\le r\le a(1+\cos t)$, cu $a>0$;

i) $\iint_D (4x-3-x^2-y^2)\,\mathrm{d}x\,\mathrm{d}y$, unde D este definit de $x^2+y^2-4x-3\le 0$;

j) $\iint_D \sqrt{a^2-x^2-y^2}\,\mathrm{d}x\,\mathrm{d}y$, unde D este definit de $x^2+y^2\le ax$, cu $a>0$;

k) $\iint_D \left(x^2+y^2-\dfrac{x+y}{2}\right)\mathrm{d}x\,\mathrm{d}y$, unde D este definit de $x^2+y^2\le 1$;

l) $\iint_D y\,\mathrm{d}x\,\mathrm{d}y$, unde $D=\left\{(x,y)\in\mathbb{R}^2 \mid x^2+y^2\le 2x+2y-1\right\}$.

11. Să se calculeze ariile domeniilor limitate de curbele:

a) $(x^2+y^2)^2=2a^2xy$;

b) $(x^2+y^2)^2=2ax^3$, cu $a>0$;

c) $(x^2+y^2)^2=a^2\left(x^2-y^2\right)$, cu $a>0$.

12. Să se calculeze următoarele integrale triple:

a) $\iiint_D \dfrac{\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z}{(1+x+y+z)^3}$, unde D este definit de $x+y+z\le 1$, $x,y,z\ge 0$;

b) $\iiint_D xyz\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$, unde D este definit de $x^2+y^2+z^2\le 1$, $x,y,z\ge 0$;

c) $\iiint_D (x^2+y^2+z^2)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$, unde D este definit de $x^2+y^2+z^2\le 1$, $x,y,z\ge 0$;

d) $\iiint_D \sqrt{R^2-x^2-y^2-z^2}\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$, unde D este definit de $x^2+y^2+z^2\le R^2$;

e) $\iiint_D (x^2+y^2-z^2)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$, unde D este definit de $x^2+y^2+z^2\le R^2$;

f) $\iiint_D \sqrt{x^2+y^2+z^2}\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$, unde D este definit de $x^2+y^2+z^2\le z$;

g) $\iiint_D ze^{-(x^2+y^2+z^2)}\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$, unde D este definit de $x^2+y^2+z^2\le a^2$, $z\ge 0$, cu $a>0$;

h) $\iiint_D \sqrt{x^2+y^2+z^2}\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$, unde D este definit de $x^2+y^2+z^2\le 2ax$, cu $a>0$.

Integrală triplă

13. Să se calculeze volumele domeniilor limitate de suprafeţele:

a) $x^2+y^2+z^2=8$ şi $x^2+y^2-2z=0$;

b) $z=4-x^2-y^2$ şi $2z=5+x^2+y^2$;

c) $2x^2+y^2+z^2=1$ şi $2x^2+y^2=z^2$;

d) $x^2+y^2+z^2=4a^2$ şi $x^2+y^2=2ax$, cu $a>0$;

e) $\frac{x^2}{4}+\frac{y^2}{9}=2z$ şi $\frac{x^2}{4}+\frac{y^2}{9}=z^2$;

f) $x^2+y^2=1$ şi $z^2=x^2+y^2$, cu $z\ge 0$.

14. Să se calculeze volumul domeniului mărginit de suprafaţa $(x^2+y^2+z^2)^2=a^2(x^2+y^2)$, cu $a>0$.

15. Să se calculeze aria paraboloidului $z = x^2 + y^2$, cu $z \in [0, 9]$.

16. Să se calculeze aria suprafeţei $x^2+y^2+z^2 = a^2$, situată în interiorul cilindrului $x^2 + y^2 = ax$, unde $a > 0$.

17. Să se calculeze aria suprafeţei $z = \sqrt{x^2 + y^2}$, situată în interiorul cilindrului $x^2 + y^2 = 2x$.

18. Să se calculeze aria decupată de $(x^2+y^2)^2 = a^2(x^2-y^2)$ pe suprafaţa $x^2+y^2 = 2az$, cu $a > 0$.

19. Să se calculeze aria decupată de $(x^2+y^2)^2 = a^2(x^2-y^2)$ pe sfera $x^2+y^2+z^2 = a^2$.

20. Să se calculeze ariile suprafeţelor:

a) $S = \left\{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 4z, z \geq x^2 + y^2\right\}$;

b) $S = \left\{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = a^2, z \geq \sqrt{x^2 + y^2}\right\}$, $a > 0$;

c) $S = \left\{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 25, 2x^2 + y^2 \leq 25\right\}$;

d) $S = \left\{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 - 4z^2 \leq 0, x^2 + y^2 - 4x \leq 0\right\}$.

21. Să se calculeze fluxul câmpului vectorial $\overline{V}(x, y, z) = x\,\overline{i} + y\,\overline{j} + z\,\overline{k}$ prin faţa exterioară a conului $z^2 = x^2 + y^2$, cu $z \in [0, 1]$.

22. Să se calculeze fluxul câmpului vectorial $\overline{V}(x, y, z) = y\,\overline{i} - x\,\overline{j} + z^2\,\overline{k}$ prin faţa exterioară a paraboloidului $z = x^2 + y^2$, cu $z \in [0, 1]$.

23. Să se calculeze fluxul câmpului vectorial $\overline{V}(x, y, z) = \frac{1}{\sqrt{x^2+y^2}}\left(y\,\overline{i} - x\,\overline{j} + \overline{k}\right)$ prin faţa exterioară a paraboloidului $z = 4 - x^2 - y^2$, cu $z \in [0, 1]$.

24. Să se calculeze fluxul câmpului vectorial $\overline{V}(x, y, z) = x^2\,\overline{i} + y^2\,\overline{j} + z^2\,\overline{k}$ prin faţa exterioară a sferei $x^2 + y^2 + z^2 = 1$, cu $x, y, z > 0$.

25. Să se calculeze, cu ajutorul formulei Riemann-Green, valoarea integralei

$$\int_C \left(xe^{xy} + 2x\cos y\right)\,\mathrm{d}x + \left(ye^{xy} - x^2\sin y\right)\,\mathrm{d}y,$$

unde C este frontiera domeniului $D = \left\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq a^2\right\}$, cu $a > 0$.

26. Să se calculeze, cu ajutorul formulei Riemann-Green, valoarea integralei

$$\int_C \left(x^2 + y^2\right)\,\mathrm{d}x + 3xy^2\,\mathrm{d}y,$$

unde C este frontiera domeniului $D = \left\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq ax\right\}$, cu $a > 0$.

27. Să se calculeze, cu ajutorul formulei Riemann-Green, circulaţia câmpului vectorial $\overline{V}(x, y) = xy\,\overline{i} + \frac{x^2}{2}\,\overline{j}$ de-a lungul curbei C, unde

$$C = \left\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1, x \leq 0 \leq y\right\} \cup \left\{(x, y) \in \mathbb{R}^2 \mid x + y = -1, x, y \leq 0\right\}.$$

28. Să se calculeze, cu ajutorul formulei Riemann-Green, circulaţia câmpului vectorial $\overline{V}(x, y) = e^x\cos y\,\overline{i} - e^x\sin y\,\overline{j}$ de-a lungul curbei C, unde C este o curbă arbitrară conţinută în semiplanul superior, care uneşte punctele $A(1, 0)$ şi $B(-1, 0)$, sensul fiind de la A către B.

29. Considerăm forma diferenţială $\omega = -\frac{y}{x^2+y^2}\,\mathrm{d}x + \frac{x}{x^2+y^2}\,\mathrm{d}y$.

a) Să se calculeze $\int_{\mathscr{C}(O,R)} \omega$, unde $\mathscr{C}(O,R)$ este cercul de centru $O(0,0)$ şi rază $R>0$.

b) Să se calculeze $\int_C \omega$, unde C este o curbă închisă astfel încât $O(0,0)\notin C$.

30. Considerăm forma diferenţială $\omega = \frac{x-y}{x^2+y^2}\,\mathrm{d}x + \frac{x+y}{x^2+y^2}\,\mathrm{d}y$. Să se calculeze integrala $\int_C \omega$, unde C este o curbă închisă astfel încât $O(0,0)\notin C$.

31. Să se calculeze, cu ajutorul formulei Gauss-Ostrogradski, fluxul câmpului vectorial $\overline{V}(x,y,z) = x^3\,\overline{i} + y^3\,\overline{j} + z^3\,\overline{k}$ prin faţa exterioară a sferei $x^2+y^2+z^2=a^2$, cu $a>0$.

32. Să se calculeze, cu ajutorul formulei Gauss-Ostrogradski, fluxul câmpului vectorial $\overline{V}(x,y,z) = x^3\,\overline{i} + x^2y\,\overline{j} + x^2z\,\overline{k}$ prin suprafaţa cilindrului $x^2+y^2=r^2$, cuprinsă între planele $z=0$ şi $z=a$, cu $r,a>0$.

33. Să se calculeze, cu ajutorul formulei Gauss-Ostrogradski, fluxul câmpului vectorial $\overline{V}(x,y,z) = x^2yz\,\overline{i}+xy^2z\,\overline{j}+xyz^2\,\overline{k}$ prin faţa exterioară a sferei $x^2+y^2+z^2=1$, cu $x,y,z\geq 0$.

34. Să se calculeze, cu ajutorul formulei Gauss-Ostrogradski, fluxul câmpului vectorial $\overline{V}(x,y,z) = xz\,\overline{i} + xy\,\overline{j} + yz\,\overline{k}$ prin suprafaţa S, bordul domeniului $D = \{(x,y,z)\in\mathbb{R}^3 \mid 2x^2+y^2\leq z\leq 1-(x^2+2y^2)\}$, orientată după normala interioară.

35. Să se calculeze, cu ajutorul formulei Gauss-Ostrogradski, fluxul câmpului vectorial $\overline{V}(x,y,z) = x(y^2-z^2)\,\overline{i}+y(z^2-x^2)\,\overline{j}+z(x^2-y^2)\,\overline{k}$ prin suprafaţa S, bordul domeniului $D=\{(x,y,z)\in\mathbb{R}^3 \mid x^2+y^2+z^2=a^2\}$, cu $a>0$, orientată după normala exterioară.

36. Să se calculeze, cu ajutorul formulei Gauss-Ostrogradski, fluxul câmpului vectorial $\overline{V}(x,y,z) = x(z+3)\,\overline{i}+yz\,\overline{j}-(z+z^2)\,\overline{k}$ prin suprafaţa sferei $x^2+y^2+z^2=1$, cu $z\geq 0$.

37. Să se calculeze, cu ajutorul formulei Gauss-Ostrogradski, fluxul câmpului vectorial $\overline{V}(x,y,z) = x^2(y-z)\,\overline{i} + y^2(z-x)\,\overline{j} + z^2(x-y)\,\overline{k}$ prin suprafaţa conului $z^2=x^2+y^2$, cu $0<z\leq 1$.

38. Să se calculeze, cu ajutorul formulei lui Stokes, valoarea integralei $\int_C y^2\,\mathrm{d}x + z^2\,\mathrm{d}y + x^2\,\mathrm{d}z$, C fiind curba de intersecţie a suprafeţelor $x^2+y^2+z^2=a^2$ şi $x^2+y^2=ax$, cu $a>0$.

39. Să se calculeze, cu ajutorul formulei lui Stokes, valoarea integralei $\int_C (y-2z)\,\mathrm{d}x + (x-z)\,\mathrm{d}y + (2x-y)\,\mathrm{d}z$, C fiind curba de intersecţie a suprafeţelor $x^2+y^2+z^2=a^2$ şi $x-y+z=0$, cu $a>0$.

40. Să se calculeze, cu ajutorul formulei lui Stokes, valoarea integralei $\int_C yz^2\,\mathrm{d}x + xy\,\mathrm{d}y + x\,\mathrm{d}z$, C fiind curba de intersecţie a suprafeţelor $x^2+y^2=4z$ şi $x+y+z=1$.

41. Să se calculeze, cu ajutorul formulei lui Stokes, valoarea integralei $\int_C (x^2+xy-1)\,\mathrm{d}x + (y+z)\,\mathrm{d}y + (xz^2-yz)\,\mathrm{d}z$, C fiind curba de intersecţie a suprafeţelor $\frac{x^2}{4}+y^2+z^2=1$ şi $y+z=0$.

42. **(Legea lui Gauss)** Pentru orice $q>0$ considerăm câmpul scalar $f(x,y,z)=\frac{q}{4\pi\sqrt{x^2+y^2+z^2}}$ şi câmpul de gradienţi $\overline{V}=-\operatorname{grad} f$. Să se arate că fluxul câmpului $\overline{V}$ prin orice suprafaţă închisă ce conţine originea în interior este egal cu q.

43. Să se calculeze fluxul câmpului vectorial $\overline{V}=\frac{1}{r}\left(\overline{r}\times\overline{k}\right)$:
a) printr-o suprafaţă închisă arbitrară ce nu conţine originea în interior;
b) prin sfera unitate.

44. Să se calculeze fluxul câmpului vectorial $\overline{V}=\overline{r}+\frac{\overline{k}\cdot\overline{r}}{r^4}\overline{r}$ prin suprafaţa $S=\left\{(x,y,z)\in\mathbb{R}^3 \mid z=3-x^2-y^2, z\geq 1\right\}\cup\left\{(x,y,z)\in\mathbb{R}^3 \mid x^2+y^2\leq 2, z=1\right\}$, orientată după normala exterioară.

Răspunsuri

1. a) $\frac{\pi\sqrt{2}}{4}$; b) $\frac{\pi\sqrt{3}}{3}$; c) $\frac{\pi}{3\sin\frac{\pi}{9}}$; d) $\frac{\pi\sqrt{2}}{4}$; e) $\frac{1}{q}\Gamma\left(\frac{p+1}{q}\right)$; f) $\frac{\sqrt{\pi}}{2}$; g) $\frac{3\pi}{512}$; h) $\Gamma(p+1)$. **2.** a) $\pi\ln\frac{y+1}{2}$; b) $\pi\arcsin y$; c) $\frac{\pi}{2}\ln(y+1)$; d) $\pi\left(\sqrt{1-a^2}-1\right)$. **3.** a) $8a$; b) $16a$; c) $\frac{a\sqrt{1+m^2}}{m}\left(e^{2\pi m}-1\right)$. **4.** a) $\frac{\pi^2}{16}-\ln\sqrt{2}$; b) $\frac{2\sqrt{2}}{3}(b^3-a^3)$; c) 0; d) $\frac{2\sqrt{2}a^3}{3}$; e) $\frac{2\sqrt{2}}{3}\left((1+2\pi^2)^{\frac{3}{2}}-1\right)$; f) $\frac{\pi}{4a^5}$; g) $\frac{\pi}{4a^5}$. **5.** a) $-2\pi a^2$; b) $\frac{3\pi a^{\frac{4}{3}}}{16}$; c) $4\pi a^3$; d) $4\pi a^3$; e) $-\frac{a^4}{12}$; f) -8π; g) $\frac{a^2b^2}{20}\left(e^{-8\pi}-1\right)$; h) -4π. **7.** 1. **8.** $\ln 5$. **10.** a) $\pi(e-1)$; b) $\frac{\pi}{4}\left(\frac{\pi}{2}-1\right)$; c) $\frac{a^2b^2}{3(a+b)}$; d) $\frac{\pi ab}{e}$; e) $\pi\operatorname{arctg} a+\frac{\pi a}{1+a^2}$; f) $\frac{7}{45}$; g) $\frac{\pi}{8}+\frac{2}{9}$; h) $\frac{\pi a^3}{2}$; i) $\frac{\pi}{2}$; j) $\frac{2a^3}{3}\left(\frac{\pi}{2}-\frac{2}{3}\right)$; k) $\frac{\pi}{2}$; l) π. **11.** a) a^2; b) $\frac{5a^2\pi}{8}$; c) a^2. **12.** a) $\frac{1}{2}\left(\ln 2-\frac{5}{8}\right)$; b) $\frac{1}{2}\left(\ln 2-\frac{5}{8}\right)$; c) $\frac{1}{2}\left(\ln 2-\frac{5}{8}\right)$; d) $\frac{\pi^2R^4}{4}$; e) $\frac{4\pi R^5}{15}$; f) $\frac{\pi}{10}$; g) $\frac{\pi}{2}\left(1-(1+a^2)e^{-a^2}\right)$; h) $\frac{8\pi}{5}a^4$. **13.** a) $\frac{4\pi}{3}(8\sqrt{2}-7)$; b) $\frac{3}{4}\pi$; c) $\frac{\pi}{3}(\sqrt{2}-1)$; d) $\frac{16a^3}{9}(3\pi-4)$; e) 8π; f) $\frac{\pi}{3}$. **14.** $\frac{\pi^2a^3}{4}$. **15.** $\frac{\pi}{6}\left(\sqrt{37^3}-1\right)$. **16.** $4a^2$. **17.** $\pi\sqrt{2}$. **18.** $\frac{1}{9}a^2(20-3\pi)$. **19.** $2a^2(\pi+4-4\sqrt{2})$. **20.** a) 4π; b) $\pi a^2(2-\sqrt{2})$; c) 100π; d) $4\pi\sqrt{5}$. **21.** 0. **22.** $\frac{\pi}{3}$. **23.** $2\pi(2-\sqrt{3})$. **24.** $\frac{3\pi}{8}$. **25.** 0. **26.** $\frac{3\pi a^4}{64}$. **27.** 0. **28.** $1-e$. **29.** a) 2π; b) 2π. **31.** $\frac{12\pi a^5}{5}$. **32.** $\frac{5\pi ar^4}{4}$. **33.** $\frac{1}{8}$. **34.** $-\frac{\pi}{12}$. **35.** 0. **36.** $\frac{4\pi^3}{3}$. **37.** 0. **38.** $-\frac{\pi a^3}{4}$. **39.** $\frac{16\pi a^2\sqrt{3}}{9}$. **40.** -720π. **41.** $-\frac{\pi\sqrt{2}}{8}$.

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REZOLVĂRI

rezolvări

1. Să se calculeze următoarele **integrale improprii**:

a) $\int_0^\infty \frac{dx}{1+x^4}$

Facem schimbarea de variabilă $t = x^4 \Rightarrow x = t^{\frac{1}{4}} \Rightarrow$

$dx = \frac{1}{4} t^{-\frac{3}{4}} dt \Rightarrow I = \int_0^\infty \frac{1}{4} \cdot t^{-3/4} \cdot (1+t)^{-1} dt = B\left(\frac{1}{4}, \frac{3}{4}\right) \cdot \frac{1}{4} =$

$= \frac{\Gamma\left(\frac{1}{4}\right)\Gamma\left(\frac{3}{4}\right)}{\Gamma(1)} \cdot \frac{1}{4} = \frac{\pi}{\sin\frac{\pi}{4}} \cdot \frac{1}{4} = \frac{\pi}{\frac{\sqrt{2}}{2}} \cdot \frac{1}{4} = \frac{\pi}{2\sqrt{2}} = \frac{\pi\sqrt{2}}{4}$

funcția beta

$$B(p,q) = \frac{\Gamma(p)\cdot\Gamma(q)}{\Gamma(p+q)}$$

$$\Gamma(p)\cdot\Gamma(1-p) = \frac{\pi}{\sin p\pi}, \quad 0<p<1$$

$$\Gamma(n+1) = n!, \quad n \in \mathbb{N}$$

funcția gama

b) $\int_0^\infty \frac{dx}{\sqrt[3]{x}\,(1+x^2)}$

Facem schimbarea de variabilă $t = x^2 \Rightarrow x = t^{1/2} \Rightarrow$

$dx = \frac{1}{2} t^{-1/2} dt \Rightarrow I = \int_0^\infty \frac{1}{2} t^{-1/2} \cdot t^{-1/6} \cdot (1+t)^{-1} dt = \int_0^\infty \frac{1}{2} \cdot t^{-2/3} \cdot (1+t)^{-1} dt =$

$= \frac{1}{2} B\left(\frac{1}{3}, \frac{2}{3}\right) = \frac{1}{2} \frac{\Gamma\left(\frac{1}{3}\right)\cdot\Gamma\left(\frac{2}{3}\right)}{\Gamma(1)} = \frac{1}{2} \cdot \frac{\pi}{\sin(\pi/3)} = \frac{1}{2} \cdot \frac{\pi}{\sqrt{3}/2} = \frac{\pi\sqrt{3}}{3}$

c) $\int_0^\infty \frac{dx}{\sqrt[3]{x^2}\,(1+x^3)}$

Facem schimbarea de variabilă $t = x^3 \Rightarrow x = t^{1/3} \Rightarrow dx = \frac{1}{3} t^{-2/3} dt$

$\Rightarrow I = \int_0^\infty \frac{1}{3} \cdot t^{-2/3} \cdot t^{-2/9} (1+t)^{-1} dt = \int_0^\infty \frac{1}{3} t^{-8/9} \cdot (1+t)^{-1} dt = \frac{1}{3} B\left(\frac{1}{9}, \frac{8}{9}\right)$

$= \frac{1}{3} \cdot \frac{\Gamma\left(\frac{1}{9}\right)\cdot\Gamma\left(\frac{8}{9}\right)}{\Gamma(1)} = \frac{1}{3} \cdot \frac{\pi}{\sin\frac{\pi}{9}} = \frac{\pi}{3\sin\frac{\pi}{9}}$

d) $\int_0^\infty \frac{\sqrt[4]{x}}{(1+x)^2} dx = B\left(\frac{5}{4}, \frac{3}{4}\right) = \frac{\Gamma\left(\frac{5}{4}\right)\Gamma\left(\frac{3}{4}\right)}{\Gamma(2)} = \Gamma\left(\frac{5}{4}\right)\Gamma\left(\frac{3}{4}\right) = \frac{1}{4}\Gamma\left(\frac{1}{4}\right)\cdot\Gamma\left(\frac{3}{4}\right) =$

$= \frac{1}{4} \cdot \frac{\pi}{\sin \pi/4} = \frac{1}{4} \frac{\pi}{\frac{\sqrt{2}}{2}} = \frac{\pi}{2\sqrt{2}} = \frac{\pi\sqrt{2}}{4}$

e) $\int_0^\infty x^p e^{-x^q}$, $p>-1$, $q>0$

Facem schimbarea de variabilă $t = x^q \Rightarrow$

$x = t^{1/q} \Rightarrow dx = \frac{1}{q} \cdot t^{\frac{1-q}{q}} dt \Rightarrow$

$$I = \int_0^\infty \frac{1}{q} \cdot t^{\frac{1-q}{q}} \cdot t^{p/q} \cdot e^{-t} dt = \int_0^\infty \frac{1}{q} \cdot t^{\frac{1+p-q}{q}} \cdot e^{-t} dt =$$

$$= \Gamma\left(\frac{1+p-q}{q}+1\right) \cdot \frac{1}{q} = \Gamma\left(\frac{1+p}{q}\right) \cdot \frac{1}{q}$$

f) $\int_0^\infty e^{-x^2} dx$

Facem schimbarea de variabilă $t = x^2 \Leftrightarrow$

$x = t^{1/2} \Rightarrow dx = \frac{1}{2} t^{-1/2} dt \Rightarrow$

$$I = \int_0^\infty \frac{1}{2} \cdot t^{-\frac{1}{2}} \cdot e^{-t} dt = \Gamma\left(\frac{1}{2}\right) \cdot \frac{1}{2} = \frac{\sqrt{\pi}}{2}$$

g) $\int_0^{\pi/2} \sin^6 x \cos^4 x \, dx$

Facem schimbarea de variabilă $y = \sin^2 x \Leftrightarrow$

$dy = 2\sin x \cos x \, dx \Rightarrow$

$$I = \int_0^1 y^3 \cdot (1-y)^2 \cdot \frac{1}{2} \cdot \frac{1}{\sqrt{y} \cdot \sqrt{1-y}} dy = \frac{1}{2} \int_0^1 y^{\frac{5}{2}} \cdot (1-y)^{\frac{3}{2}} dy$$

$$= \frac{1}{2} B\left(\frac{5}{2}+1, \frac{3}{2}+1\right) = \frac{1}{2} B\left(\frac{7}{2}, \frac{5}{2}\right) = \frac{1}{2} \frac{\Gamma\left(\frac{7}{2}\right) \cdot \Gamma\left(\frac{5}{2}\right)}{\Gamma\left(\frac{12}{2}\right)} = \frac{1}{2} \frac{\frac{5}{2} \Gamma\left(\frac{5}{2}\right) \cdot \frac{3}{2} \cdot \Gamma\left(\frac{3}{2}\right)}{\Gamma(6)}$$

$$= \frac{1}{2} \cdot \frac{\frac{15}{4} \Gamma\left(\frac{3}{2}\right) \cdot \frac{3}{4} \Gamma\left(\frac{1}{2}\right)}{\Gamma(6)} = \frac{1}{2} \cdot \frac{\frac{15}{4} \cdot \frac{3}{4} \cdot \frac{1}{2} \Gamma^2\left(\frac{1}{2}\right)}{\Gamma(6)} = \frac{45}{32} \cdot \frac{\pi}{5!} \cdot \frac{1}{2} =$$

$$= \frac{1}{2} \cdot \frac{45}{32} \cdot \frac{\pi}{120} = \frac{3\pi}{256} \cdot \frac{1}{2} = \frac{3\pi}{512}$$

h) $\int_0^1 \left(\ln \frac{1}{x}\right)^p dx$, $p>-1$

Facem schimbarea de variabilă $y = \ln\frac{1}{x} \Leftrightarrow$

$dy = \frac{1}{\frac{1}{x}} \cdot -\frac{1}{x^2}\,dx \Leftrightarrow dy = -\frac{1}{x}\,dx \Rightarrow$

$$I = -\int_{\infty}^{0} y^{\rho} \cdot e^{-y}\,dy = \int_{0}^{\infty} y^{\rho} e^{-y}\,dy = \Gamma(\rho+1)$$

(2) Integrale folosind derivarea sub semnul integral

a) $\int_0^{\pi/2} \ln(\cos^2 x + y^2 \sin^2 x)\,dx$, cu $y>0$

Notăm $I(y) = \int_0^{\pi/2} \ln(\cos^2 x + y^2 \sin^2 x)\,dx$, $y>0$

$\Rightarrow$ conform teoremei lui Leibniz:

$$I'(y) = \int_0^{\pi/2} \frac{\partial f}{\partial y}(x,y)\,dx = \int_0^{\pi/2} \frac{2y\sin^2 x}{\cos^2 x + y^2 \sin^2 x}\,dx \overset{\text{calculez}}{=} \dots =$$

$\frac{\pi}{y+1} \Rightarrow I(y) = \int \frac{\pi}{y+1}\,dy = \pi \ln(y+1) + K$

Pentru $y=1 \Rightarrow I(1) = \int_0^{\pi/2} \ln(\cos^2 x + \sin^2 x)\,dx = \int_0^{\pi/2} 0\,dx = 0$

dar $I(1) = \pi \ln 2 + K \Rightarrow K = -\pi \ln 2 \Rightarrow$

$I(y) = \pi \ln(y+1) - \pi \ln 2 = \pi \ln\frac{y+1}{2}$

(3) Să se calculeze lungimile următoarelor curbe:

a) $C: \begin{cases} x = a(t - \sin t) \\ y = a(1 - \cos t) \end{cases}$, $t \in [0, 2\pi]$, $a>0$

Lungimea unei curbe C este:

$$\int_C ds = \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2}\,dt = \mathcal{L}(C)$$

$$\int_C ds = \int_0^{2\pi} \sqrt{[a(1-\cos t)]^2 + (a\sin t)^2}\,dt = \int_0^{2\pi} \sqrt{a^2(1 - 2\cos t + \cos^2 t) +}$$

$$\int_0^{2\pi} \sqrt{a^2\sin^2 t}\,dt = \int_0^{2\pi} \sqrt{a^2(1-2\cos t+\cos^2 t+\sin^2 t)}\,dt =$$

$$= a\int_0^{2\pi} \sqrt{2-2\cos t}\,dt = a\sqrt{2}\int_0^{2\pi} \sqrt{1-\cos t}\,dt = a\sqrt{2}\int_0^{2\pi} \sqrt{2\sin^2\frac{t}{2}}\,dt =$$

$$= a\sqrt{2}\cdot\sqrt{2}\int_0^{2\pi} \left|\sin\frac{t}{2}\right|dt = 2a\int_0^{2\pi} \sin\frac{t}{2}\,dt = 4a\left(-\cos\frac{t}{2}\right)\Big|_0^{2\pi}$$

$$= 4a(-\cos\pi+\cos 0) = 8a$$

b) $C:\begin{cases} x = a(2\cos t-\cos 2t) \\ y = a(2\sin t-\sin 2t)\end{cases}$, $t\in[0,2\pi]$, $a>0$

Lungimea curbei C este $\int_C ds$.

$$\int_C ds = \int_0^{2\pi} \sqrt{(x'(t))^2+(y'(t))^2}\,dt$$

$$x'(t) = a(-2\sin t+2\sin 2t) = 2a(\sin 2t-\sin t)$$

$$y'(t) = a(2\cos t-2\cos 2t) = 2a(\cos t-\cos 2t)$$

$$\Rightarrow \int_C ds = \int_0^{2\pi} \sqrt{4a^2(\sin 2t-\sin t)^2+4a^2(\cos t-\cos 2t)^2}\,dt =$$

$$= \int_0^{2\pi} 2a\cdot\sqrt{\sin^2 2t-2\sin 2t\sin t+\sin^2 t+\cos^2 t-2\cos 2t\cos t+\cos^2 2t}\,dt =$$

$$= 2a\int_0^{2\pi} \sqrt{2-2\sin 2t\sin t-2\cos 2t\cos t}\,dt = 2\sqrt{2}a\int_0^{2\pi} \sqrt{1-\sin 2t\sin t-\cos 2t\cos t}\,dt$$

$$= 2\sqrt{2}a\int_0^{2\pi} \sqrt{1-\frac{1}{2}(\cos t-\cos 3t)-\frac{1}{2}(\cos t+\cos 3t)}\,dt$$

$$= 2\sqrt{2}a\int_0^{2\pi} \sqrt{1-\cos t}\,dt = 2\sqrt{2}a\int_0^{2\pi} \sqrt{2\sin^2\frac{t}{2}}\,dt = 4a\int_0^{2\pi} \sqrt{\sin^2\frac{t}{2}}\,dt =$$

$$4a\int_0^{2\pi} \sin\frac{t}{2}\,dt = 8a\left(-\cos\frac{t}{2}\right)\Big|_0^{2\pi} = 8a(1+1) = 16a$$

c) $C:\begin{cases} x = ae^{ut}\cos t \\ y = ae^{ut}\sin t\end{cases}$, $t\in[0,2\pi]$, $a,u>0$

Lungimea curbei C este $\int_C ds$.

$$\int_C ds = \int_0^{2\pi} \sqrt{(x'(t))^2 + (y(t))^2}\, dt = \int_0^{2\pi} \sqrt{(ae^{mt}\cdot(-\sin t) + ame^{mt}\cos t)^2}$$

$$+ (ae^{mt}\cos t + mae^{mt}\sin t)^2\, dt = a\int_0^{2\pi} \sqrt{e^{2mt}\sin^2 t - 2m\cdot}$$

$$e^{2mt}\sin t\cos t + m^2 e^{2mt}\cos^2 t + e^{2mt}\cos^2 t + 2me^{2mt}\sin t\cos t$$

$$+ m^2 e^{2mt}\sin^2 t\, dt = a\int_0^{2\pi} e^{mt}\sqrt{1+m^2}\, dt = \frac{a\sqrt{1+m^2}}{m}\cdot e^{mt}\Big|_0^{2\pi}$$

$$= \frac{a\sqrt{1+m^2}}{m}\left(e^{2\pi m} - 1\right)$$

4.) **Integrale curbilinii de speța I**

a) $\int_C \underbrace{ye^{-x}}_{\rho} ds$, $C: \begin{cases} x = \ln(1+t^2) \\ y = 2\operatorname{arctg} t - t \end{cases}$, $t \in [0,1]$

$$\Rightarrow \int_C ye^{-x} ds = \int_0^1 (2\operatorname{arctg} t - t)\, e^{-\ln(1+t^2)} \cdot \sqrt{(x'(t))^2 + (y(t))^2}\, dt =$$

$$= \int_0^1 (2\operatorname{arctg} t - t)\cdot \frac{1}{1+t^2}\cdot \sqrt{\frac{4t^2}{(1+t^2)^2} + \left(2\cdot\frac{1}{1+t^2} - 1\right)^2}\, dt =$$

$$\int_0^1 (2\operatorname{arctg} t - t)\cdot \frac{1}{1+t^2}\cdot \sqrt{\frac{4t^2}{(1+t^2)^2} + \frac{(2-1-t^2)^2}{(1+t^2)^2}}\, dt =$$

$$= \int_0^1 (2\operatorname{arctg} t - t)\cdot \frac{1}{1+t^2}\cdot \sqrt{\frac{4t^2+1-2t^2+t^4}{(1+t^2)^2}}\, dt =$$

$$= \int_0^1 (2\operatorname{arctg} t - t)\cdot \frac{1}{(1+t^2)}\cdot \sqrt{\frac{(t^2+1)^2}{(1+t^2)^2}}\, dt = \int_0^1 (2\operatorname{arctg} t - t)\cdot \frac{1}{1+t^2}\, dt$$

$$= \left(\operatorname{arctg}^2 t - \frac{1}{2}\ln(1+t^2)\right)\Big|_0^1 = \frac{\pi^2}{16} - \ln\sqrt{2}$$

b) $\int_C (x^2+y^2)\,ds$, C este segmentul $[AB]$, cu $A(a,a)$, $B(b,b)$, $b>a$.

Parametrizare segment $[AB]$: $\begin{cases} x=t \\ y=t \end{cases}$, $t\in[a,b]$

$$\Rightarrow \int_C (x^2+y^2)\,ds = \int_a^b (t^2+t^2)\cdot\sqrt{1^2+1^2}\,dt = 2\sqrt{2}\int_a^b t^2\,dt =$$

$$\frac{2\sqrt{2}}{3}t^3\Big|_a^b = \frac{2\sqrt{2}}{3}(b^3-a^3)$$

c) $\int_C xy\,ds$, $C:\begin{cases} x=a\cos^2 t \\ y=a\cos t \end{cases}$, $t\in[0,\pi]$, $a>0$.

$$\int_C xy\,ds = \int_0^{\pi} a\cos^2 t\cdot a\cos t\cdot\sqrt{(2a\cos t\sin t)^2+a^2\cos^2 t}\,dt =$$

$$= \int_0^{\pi} a^2\cos^3 t\cdot\sqrt{4a^2\cos^2 t\sin^2 t+a^2\cos^2 t}\,dt = a^3\int_0^{\pi}\cos^3 t\,|\cos t|\cdot$$

$$\sqrt{1+4\sin^2 t}\,dt = a^3\int_0^{\pi/2}\cos^4 t\cdot\sqrt{1+4\sin^2 t}\,dt + a^3\int_{\pi/2}^{\pi}-\cos^4 t\sqrt{1+4\sin^2 t}\,dt$$

$$= a^3\int_0^{\pi/2}\cos^4 t\cdot\sqrt{1+4\sin^2 t}\,dt + a^3\int_{\pi-\pi/2}^{\pi-\pi}\cos^4 t\sqrt{1+4\sin^2 t}\,dt =$$

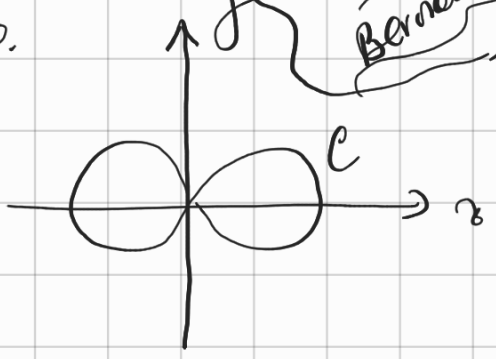
$$= a^3\int_0^{\pi/2}\cos^4 t\sqrt{1+4\sin^2 t}\,dt + a^3\int_0^{\pi/2}\cos^4 t\sqrt{1+4\sin^2 t}\,dt = 0$$

am făcut schimbare de variabilă $t \to \pi - t$

(*) d) $\int_C x\sqrt{x^2-y^2}\,ds$, C este definită de

$$(x^2+y^2)^2 = a^2(x^2-y^2),\ x\geq 0,\ a>0.$$

Lemniscata lui Bernoulli



Parametrizare pt. înfiruirea asta

$$x = \frac{a\cos t}{1+\sin^2 t} \qquad y = \frac{a\sin t\cos t}{1+\sin^2 t}$$

$\Rightarrow$ - - - -

e) $\int_C (2z - \sqrt{x^2+y^2})\,ds$, unde $C: \begin{cases} x = t\cos t \\ y = t\sin t \\ z = t \end{cases}$, $t \in [0, 2\pi]$

$$\int_C (2z - \sqrt{x^2+y^2})\,ds = \int_0^{2\pi} (2t - \sqrt{t^2\cos^2 t + t^2\sin^2 t}) \cdot \sqrt{(\cos t - \sin t + t\cos t)^2 + (\sin t + t\cos t)^2 + 1}\,dt =$$

$$= \int_0^{2\pi} (2t - t) \cdot \sqrt{\cos^2 t - 2t\sin t\cos t + \sin^2 t + t^2 + \sin^2 t + 2t\sin t\cos t + t^2\cos^2 t + 1}\,dt =$$

$$= \int_0^{2\pi} t \cdot \sqrt{1 + t^2 + 1}\,dt = \int_0^{2\pi} t\sqrt{t^2+2}\,dt = (t^2+2)^{\frac{3}{2}} \cdot \frac{2}{3} \cdot \frac{1}{2}\Big|_0^{2\pi}$$

$$= \frac{1}{3}\cdot(t^2+2)^{\frac{3}{2}} = \frac{\sqrt{(4\pi^2+2)^3} - 2\sqrt{2}}{3} = \frac{2\sqrt{2}}{3}\left((1+2\pi^2)^{\frac{3}{2}} - 1\right)$$

f) $\int_C \frac{ds}{(x^2+y^2+z^2)^3}$, $C: x^2+y^2+z^2 = 2a^2$, $z = a > 0$.

$z = a > 0 \Rightarrow x^2 + y^2 = a^2 \Rightarrow$ C este cercul de rază a și centru în $(0,0,a) \Rightarrow$

coordonate polare

$$\begin{cases} x = a\cos t \\ y = a\sin t \\ z = a \end{cases}, \quad t \in [0, 2\pi]$$

$$\Rightarrow \int_C \frac{ds}{(x^2+y^2+z^2)^3} = \int_0^{2\pi} \frac{1}{(a^2\cos^2 t + a^2\sin^2 t + a^2)^3} \cdot \sqrt{(-\sin t\, a)^2 + a^2\cos^2 t + 0^2}\,dt =$$

$$= \int_0^{2\pi} \frac{1}{8a^6} \cdot a\,dt = \frac{1}{8a^5}\cdot t\Big|_0^{2\pi} = \frac{\pi}{4a^5}$$

5. Integrale curbilinii de speța a II-a

a) $\int_C (2a-y)\,dx + x\,dy$, $C: \begin{cases} x = a(t - \sin t) \\ y = a(1-\cos t) \end{cases}$, $t \in [0, 2\pi]$

$$\int_C (2a-y)\,dx + x\,dy = \int_0^{2\pi} [(2a - a + a\cos t)\cdot a(1-\cos t) + a(t-\sin t)\cdot a\cdot \sin t]\,dt = \int_0^{2\pi} a^2(1-\cos^2 t) + a^2(t\sin t - \sin^2 t)\,dt$$

$$= a^2 \int_0^{2\pi} (1 - \cos^2 t + t\sin t - \sin^2 t)\,dt = a^2 \int_0^{2\pi} t\sin t\,dt = -a^2 t\cos t \Big|_0^{2\pi} +$$

$$a^2 \int_0^{2\pi} \cos t\,dt = -a^2 \cdot 2\pi + a^2 \cdot -\sin t \Big|_0^{2\pi} = -2\pi a^2$$

Model subiect 4 examen AA

pr 7 ex. Negrescu

carte Olteanu

6. Să se calculeze $\int_\Gamma ydx + xdy$ pe un drum cu capetele $A(2,1)$ şi $B(1,3)$.

Soluţie

Forma diferenţială $\alpha = ydx + xdy$ este închisă pe R^2 şi deci este exactă. Rezultă că integrala este independentă de drumul particular care uneşte punctele A şi B. Integrala se calculează pe un drum particular, de exemplu pe segmentul $[AB]$, a cărui parametrizare este:

$$x(t) = \frac{5-t}{2},\ y(t) = t,\ t \in [1,3].$$

O altă metodă constă în a determina un potenţial scalar f pentru 1-forma diferenţială α:

$$f(x,y) = \int_{x_0}^{x} y_0 dx + \int_{y_0}^{y} x dy = xy + k,$$

k fiind o constantă arbitrară. Integrala cerută în enunţ este:

$$\int_\Gamma \alpha = f(B) - f(A) = 1,$$

Γ fiind un drum arbitrar având capetele A şi B.

$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

(7) Forma diferențială $\omega = ydx + xdy$ este închisă pe $\mathbb{R}^2$ și deci este exactă. $\Rightarrow$ integrala este independentă de drumul particular care unește punctele A și B. Integrala se calculează pe un drum particular, de exemplu pe segmentul [AB] a cărui parametrizare este:

$$\begin{cases} x(t) = t \\ y(t) = \frac{x-2}{1-2} \cdot (3-1) + 1 = (2-t)\cdot 2 + 1 = 4 - 2t + 1 = 5 - 2t \end{cases}$$

segmentul [BA]

$t \in [1,2]$ (am luat prima parametrizare a acestei)

$$\Rightarrow \int_C y\,dx + x\,dy = -\int_1^2 [5-2t+t(-2)]\,dt = -\int_1^2 (-4t+5)\,dt =$$

$$= (-2t^2+5t)\Big|_1^2 = -(-8+10+2-5) = 1$$

(10.) **Integrale duble**

a) $\iint_\Delta e^{x^2+y^2}\,dx\,dy$, $\Delta = \{(x,y)\in\mathbb{R}^2 \mid x^2+y^2\le 1\}$

Parametrizare Δ: $\begin{cases} x = \rho\cos t \\ y = \rho\sin t \end{cases}$ $t\in[0,2\pi]$, $\rho\in[0,1]$ Jacobian: ρ

$$\Rightarrow \iint_\Delta e^{x^2+y^2}\,dx\,dy = \int_0^{2\pi} dt \int_0^1 e^{\rho^2}\cdot\rho\,d\rho = \int_0^{2\pi} \frac{1}{2}e^{\rho^2}\Big|_0^1 dt =$$

$$= \frac{1}{2}\int_0^{2\pi}(e-1)\,dt = \frac{1}{2}(e-1)t\Big|_0^{2\pi} = (e-1)\pi$$

c) $\iint_\Delta \frac{xy\,dx\,dy}{\sqrt{x^2+y^2}}$, unde $\Delta = \{(x,y)\in\mathbb{R}^2 \mid \frac{x^2}{a^2}+\frac{y^2}{b^2}\le 1,$

$x,y\ge 0\}$

Parametrizare Δ: $\begin{cases} x = a\rho\cos t \\ y = b\rho\sin t \end{cases}$, $\rho\in[0,1]$, $t\in[0,\frac{\pi}{2}]$ Jacobian: $ab\rho$

$$\Rightarrow \iint_\Delta \frac{xy\,dx\,dy}{\sqrt{x^2+y^2}} = \int_0^{\pi/2} dt \int_0^1 \frac{ab\rho^2\cos t\sin t}{\sqrt{a^2\rho^2\cos^2 t + b^2\rho^2\sin^2 t}}\cdot ab\rho\,d\rho$$

$$= \int_0^{\pi/2} dt \int_0^1 \frac{a^2b^2\rho^2\cos t\sin t}{\sqrt{a^2\cos^2 t+b^2\sin^2 t}}\,dt = \int_0^{\pi/2} \frac{a^2b^2\cos t\sin t}{\sqrt{a^2\cos^2 t+b^2\sin^2 t}}\,\frac{\rho^3}{3}\Big|_0^1 dt =$$

$$= \frac{1}{3}\int_0^{\pi/2} \frac{a^2b^2\cos t\sin t}{\sqrt{a^2\cos^2 t+b^2\sin^2 t}}\,dt = \frac{1}{3}\int_0^{\pi/2} \frac{a^2b^2}{b^2-a^2}\cdot(\sqrt{a^2\cos^2 t+b^2\sin^2 t})'\,dt =$$

$$= \frac{1}{3}\frac{a^2b^2}{b^2-a^2}\cdot\sqrt{a^2\cos^2 t+b^2\sin^2 t}\Big|_0^{\pi/2} = \frac{1}{3}\frac{a^2b^2}{b^2-a^2}\cdot b - \frac{1}{3}\frac{a^2b}{b^2-a^2}\cdot a =$$

$= \frac{1}{3}\frac{a^2b^2}{(b-a)(a+b)}(b-a) = \frac{1}{3}\frac{a^2b^2}{a+b}$

g) $\iint_D (1+\sqrt{x^2+y^2})\,dxdy$, $D = \{(x,y)\in\mathbb{R}^2 \mid x^2+y^2 \le y, x\ge 0\}$

Parametrizare D $\quad \rho \neq 0$

$\begin{cases} x = \rho\cos t \\ y = \rho\sin t \end{cases}$ $\quad \rho^2 \le \rho\sin t \Rightarrow \rho \le \sin t \Rightarrow \begin{cases} \rho\in[0,\sin t] \\ t\in[0,\frac{\pi}{2}] \end{cases}$

Jacobian ρ

$\Rightarrow \int_0^{\pi/2} dt \int_0^{\sin t} (1+\rho)\cdot\rho\,d\rho = \int_0^{\pi/2} \left(\frac{\rho^3}{3}+\frac{\rho^2}{2}\right)\Big|_0^{\sin t} dt =$

$= \int_0^{\pi/2} \left(\frac{\sin^3 t}{3} + \frac{\sin^2 t}{2}\right) dt = \overset{\text{calcule}}{\cdots} = \frac{\pi}{8}+\frac{2}{9}$

(11) Ariile domeniilor limitate de curbe

a) $(x^2+y^2)^2 = 2a^2xy$

$\Delta: (x^2+y^2)^2 \le 2a^2xy$

Jacobian ρ

$\begin{cases} x = \rho\cos t \\ y = \rho\sin t \end{cases} \Rightarrow \rho^4 \le 2a^2\rho^2\cos t\sin t \Rightarrow \rho \neq 0$

$\rho^2 \le 2a^2\cos t\sin t \Rightarrow \rho^2 \le a^2\sin 2t$

$\Rightarrow \rho \le a\sqrt{\sin 2t} \Rightarrow \begin{cases} \rho\in[0, a\sqrt{\sin 2t}] \\ t\in[0,\frac{\pi}{2}] \end{cases}$

$\Rightarrow$ Aria domeniului limitat de curba dată este: $\iint_\Delta dxdy = \int_0^{\pi/2} dt \int_0^{a\sqrt{\sin 2t}} \rho\,d\rho = \int_0^{\pi/2} \frac{\rho^2}{2}\Big|_0^{a\sqrt{\sin 2t}} dt =$

$= \frac{1}{2}\int_0^{\pi/2} a^2\sin 2t\,dt = \frac{1}{4}a^2(-\cos 2t)\Big|_0^{\pi/2} = \frac{1}{2}a^2$

c) $(x^2+y^2)^2 = a^2(x^2-y^2)$, $a>0$

Parametrizare

$\begin{cases} x = \rho\cos t \\ y = \rho\sin t \end{cases}$ $\quad \rho^4 \le a^2(\rho^2\cos^2 t - \rho^2\sin^2 t) \Rightarrow \rho \neq 0$

$\rho^2 \le a^2(\cos^2 t - \sin^2 t) \Rightarrow$

$\rho^2 \le a^2\cos 2t \Rightarrow \begin{cases} \rho \in [0, a\sqrt{\cos 2t}] \\ t \in [-\frac{\pi}{2}, \frac{\pi}{2}] \end{cases}$

Jacobian ρ

$\Rightarrow$ Aria este $\iint_\Delta dx\,dy = \int_{-\pi/2}^{\pi/2} dt \int_0^{a\sqrt{\cos 2t}} \rho\, d\rho =$

$= \int_{-\pi/2}^{\pi/2} \frac{\rho^2}{2}\Big|_0^{a\sqrt{\cos 2t}} dt = \frac{1}{2}\int_{-\pi/2}^{\pi/2} a^2\cos 2t\, dt = \frac{1}{4}\sin 2t \cdot a^2\Big|_{-\pi/2}^{\pi/2} = \frac{1}{2}a^2$

(12.) a) $\iiint_\Delta \frac{dx\,dy\,dz}{(1+x+y+z)^3}$, unde Δ este definit

de $x+y+z \le 1$, $x, y, z \ge 0$.

Parametrizare

$z \le 1-x-y$

D': $x+y \le 1 \Rightarrow \Delta'$: $\begin{matrix} x+y \le 1 \\ z \in [0, 1-x-y] \end{matrix} \Rightarrow$

$\iiint_D \frac{dx\,dy\,dz}{(1+x+y+z)^3} = \iint_{\Delta'} dx\,dy \int_0^{1-x-y} \frac{1}{(1+x+y+z)^3} dz =$

$= \iint_{\Delta'} \frac{1}{(1+x+y+z)^2} \cdot \frac{-1}{2}\Big|_0^{1-x-y} dx\,dy = \iint_{\Delta'} -\frac{1}{2}\cdot\left(\frac{1}{4} - \frac{1}{(1+x+y)^2}\right) dx\,dy$

Parametrizare Δ'

$\begin{cases} x \in [0,1] \\ y \in [0, 1-t] \end{cases} \Rightarrow$

$* = \int_0^1 dx \int_0^{1-x} \left(-\frac{1}{8} + \frac{1}{2}\cdot\frac{1}{(1+x+y)^2}\right) dy = \int_0^1 \frac{-y}{8} + \frac{1}{2}\cdot\frac{-1}{1+x+y}\Big|_0^{1-x} dx$

$= \int_0^1 \left(\frac{x-1}{8} + \frac{1}{2}\cdot\frac{-1}{2} + \frac{1}{2}\cdot\frac{1}{1+x}\right) dx = \int_0^1 \left(\frac{x}{8} - \frac{3}{8} + \frac{1}{2}\cdot\frac{1}{1+x}\right) dx =$

$$= \left(\frac{x^2}{16} - \frac{3}{8}x + \frac{1}{2}\ln(1+x) \right)\Big|_0^1 = \frac{1}{16} - \frac{3}{8} + \frac{1}{2}\ln 2 = -\frac{5}{16} + \ln\sqrt{2}$$

b) $\iiint_D xyz\, dx\,dy\,dz$, $x^2+y^2+z^2 \le 1$, $x,y,z \ge 0$

Coordonate sferice

$$\begin{cases} x = \rho\sin\theta\cos\varphi \\ y = \rho\sin\theta\sin\varphi \\ z = \rho\cos\theta \end{cases} \quad \begin{cases} \rho \in [0,1] \\ \theta \in [0,\pi/2] \\ \varphi \in [0,\pi/2] \end{cases} \quad \text{Jacobian } \rho^2\sin\theta$$

$$\int_0^{\pi/2} d\varphi \int_0^{\pi/2} d\theta \int_0^1 \rho^3 \sin^2\theta \cos\varphi \sin\varphi \cos\theta\, \rho^2 \sin\theta\, d\rho =$$

$$= \int_0^{\pi/2} d\varphi \int_0^{\pi/2} \frac{\rho^5}{5} \sin^3\theta \cos\theta \cos\varphi \sin\varphi \Big|_0^1 d\theta = \frac{1}{5}\int_0^{\pi/2} d\varphi \int_0^{\pi/2} \sin^3\theta\cos\theta -$$

$$\cos\varphi\sin\varphi\, d\theta = \frac{1}{5}\int_0^{\pi/2} \frac{1}{4}\sin^4\theta \cdot \cos\varphi\cdot\sin\varphi \Big|_0^{\pi/2} d\varphi =$$

$$= \frac{1}{20}\int_0^{\pi/2} \cos\varphi\sin\varphi\, d\varphi = \frac{1}{40}\sin^2\varphi\Big|_0^{\pi/2} = \frac{1}{40}$$

f) $\iiint_\Delta -\sqrt{x^2+y^2+z^2}\, dx\,dy\,dz$, Δ: $x^2+y^2+z^2 \le z$

Parametrizare Δ

$$\begin{cases} x = \rho\sin\theta\cos\varphi \\ y = \rho\sin\theta\sin\varphi \\ z = \rho\cos\theta \end{cases} \quad \rho^2 \le \rho\cos\theta \Rightarrow \rho \le \cos\theta \Rightarrow \begin{cases} \rho \in [0,\cos\theta] \\ \theta \in [0,\frac{\pi}{2}],\ \varphi \in [0,2\pi] \end{cases}$$

Jacobian $\rho^2\sin\theta$

$$\Rightarrow \iiint_\Delta -\sqrt{x^2+y^2+z^2}\, dx\,dy\,dz = \int_0^{2\pi} d\varphi \int_0^{\pi/2} d\theta \int_0^{\cos\theta} \rho\cdot\rho^2\sin\theta\, d\rho =$$

$$= \frac{1}{4}\int_0^{2\pi} d\varphi \int_0^{\pi/2} \rho^4\sin\theta\Big|_0^{\cos\theta} d\theta = \frac{1}{4}\int_0^{2\pi} d\varphi \int_0^{\pi/2} \cos^4\theta\cdot\sin\theta\, d\theta =$$

$$= \frac{-1}{20}\int_0^{2\pi} \cos^5\theta\Big|_0^{\pi/2} d\varphi = \frac{-1}{20}\int_0^{2\pi} -1\, d\varphi = \frac{1}{20}\varphi\Big|_0^{2\pi} = \frac{\pi}{10}$$

(13) Volumul domeniilor limitate de suprafețele

a) $x^2+y^2+z^2=8$ și $x^2+y^2-2z=0$

$$\left.\begin{aligned} z^2 &= 8-x^2-y^2 \\ 2z &= x^2+y^2 \end{aligned}\right\} \Rightarrow$$

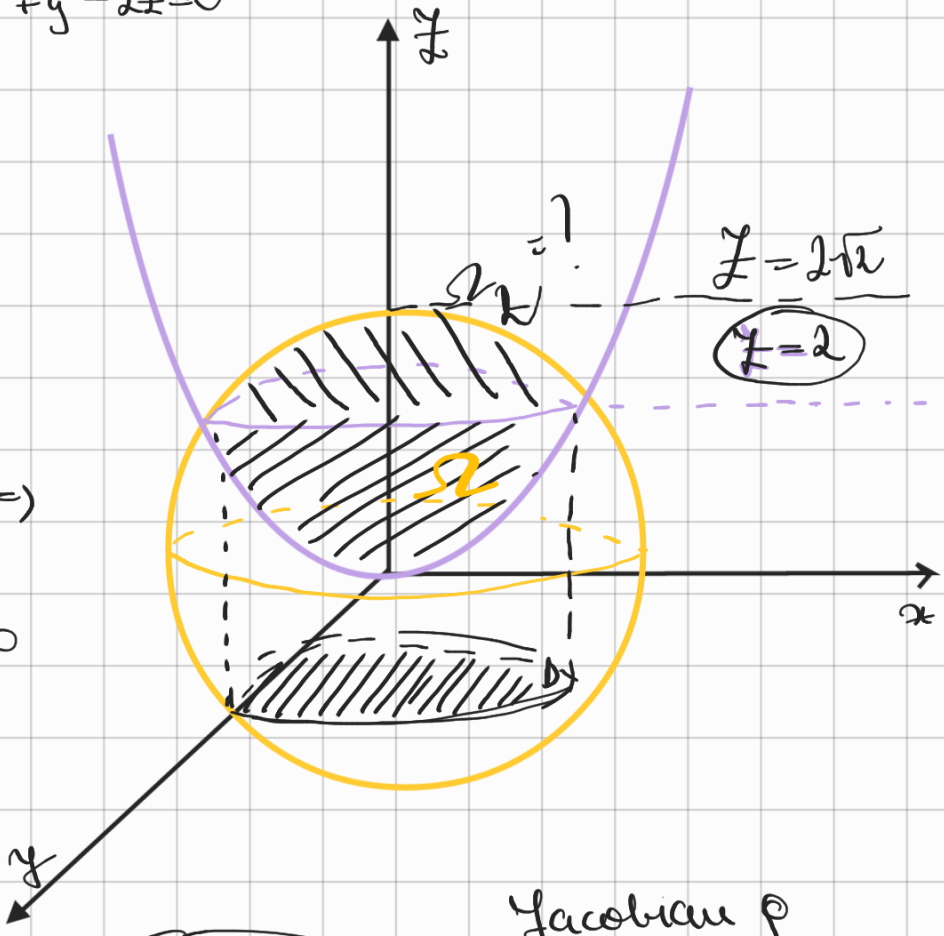
$$\sqrt{8-x^2-y^2} = \frac{x^2+y^2}{2} \Leftrightarrow$$

$$4(8-x^2-y^2) = (x^2+y^2)^2 \Leftrightarrow$$

$$(x^2+y^2)^2+4(x^2+y^2)-32=0$$

$$\Rightarrow (x^2+y^2) = \begin{cases} 4 \\ -8 \end{cases}$$

$\Rightarrow z=2$



$z \in \left[\frac{x^2+y^2}{2}, 2\right]$, $D: x^2+y^2 \le 4$ $\Rightarrow$ Jacobian ρ

$$D: \begin{cases} x=\rho\cos t \\ y=\rho\sin t \\ \rho\in[0,2] \\ t\in[0,2\pi] \end{cases}$$

$\Rightarrow$ Volumul lui Ω este:

$$\iiint_\Omega dx\,dy\,dz = \iint_D dx\,dy \int_{\frac{x^2+y^2}{2}}^{2} dz = \iint_D z\Big|_{\frac{x^2+y^2}{2}}^{2} dx\,dy =$$

$$= \iint_D \left(2-\frac{x^2+y^2}{2}\right) dx\,dy = \int_0^{2\pi} dt \int_0^2 \left(2-\frac{\rho^2}{2}\right)\rho\,d\rho =$$

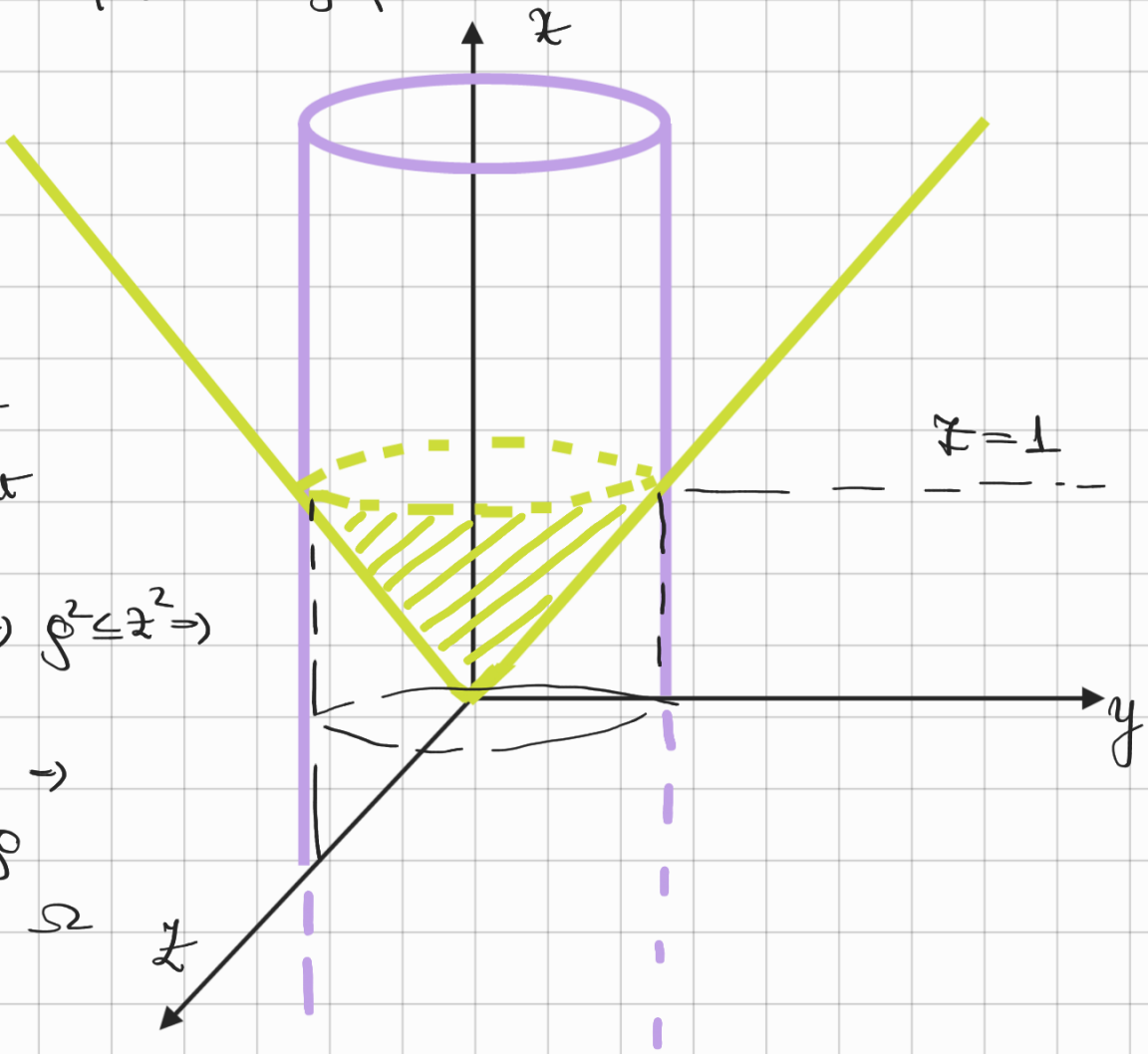
$$= \int_0^{2\pi} \left(\rho^2-\frac{\rho^4}{8}\right)\Big|_0^2 dt = \int_0^{2\pi} (4-2)\cdot dt = 4\pi$$

$$D: \begin{cases} x=\rho\cos t \\ y=\rho\sin t \\ \rho\in[0,2] \\ t\in[0,2\pi] \end{cases} \qquad z\in[2,\ 8-x^2-y^2]$$

$$\iiint_\Omega dx\,dy\,dz = \iint_D dx\,dy \int_2^{8-x^2-y^2} dz = \iint_D 6-x^2-y^2\,dx\,dy$$

$$= \int_0^{2\pi} dt \int_0^2 (6-\rho^2)\rho\,d\rho = \int_0^2 3\rho^2-\frac{\rho^4}{4}\Big|_0^2 d\rho =$$

f) $x^2+y^2=1$ și $z^2=x^2+y^2$, $z\geq 0$



Parametrizare Ω

$$\Omega:\begin{cases} z\in[0,1] \\ x=\rho\cos t \\ y=\rho\sin t \end{cases}$$

$x^2+y^2\leq z^2 \Rightarrow \rho^2\leq z^2 \Rightarrow$

$$\begin{cases} \rho\in[0,z] \\ t\in[0,2\pi] \end{cases} \Rightarrow$$

Jacobian ρ

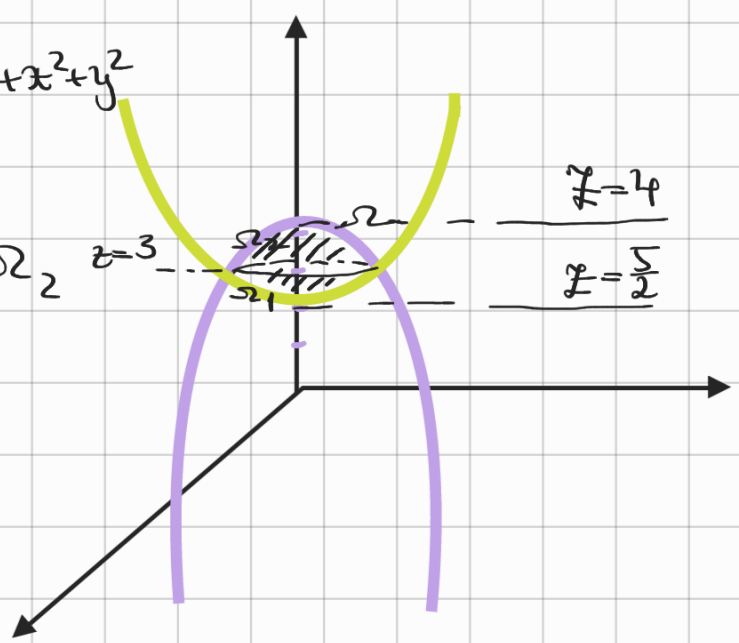
Volumul lui Ω este:

$$\iiint_\Omega dx\,dy\,dz =$$

$$= \int_0^1 dz \int_0^{2\pi} dt \int_0^z \rho\,d\rho = \int_0^1 dz \int_0^{2\pi} \frac{\rho^2}{2}\Big|_0^z dt = \int_0^1 dz \int_0^{2\pi} \frac{z^2}{2} dt =$$

$$= \int_0^1 \frac{z^2 t}{2}\Big|_0^{2\pi} dz = \int_0^1 z^2\pi\, dz = \frac{z^3\pi}{3}\Big|_0^1 = \frac{\pi}{3}$$

b) $z=4-x^2-y^2$ și $2z=5+x^2+y^2$



$$\begin{cases} z\in\left[\frac{5}{2},4\right] \\ x=\rho\cos t \\ y=\rho\sin t \end{cases} \qquad \Omega=\Omega_1+\Omega_2$$

Ω_1: $z\in\left[\frac{5}{2},3\right]$

$5+x^2+y^2\leq 2z \Rightarrow$

$5+\rho^2\leq 2z \Rightarrow$

$\rho^2 \le 2z-5 \Rightarrow \rho \in [0, \sqrt{2z-5}]$, $t \in [0, 2\pi]$ Jacobian ρ

$$\iiint_{\Omega_1} dx\,dy\,dz = \int_{5/2}^{3} dz \int_0^{2\pi} dt \int_0^{\sqrt{2z-5}} \rho\, d\rho = \int_{5/2}^{3} dz \int_0^{2\pi} \frac{\rho^2}{2}\Big|_0^{\sqrt{2z-5}} dt =$$

$$= \int_{5/2}^{3} dz \int_0^{2\pi} \frac{2z-5}{2}\, dt = \int_{5/2}^{3} \frac{2z-5}{2} \cdot t\Big|_0^{2\pi} dz = \int_{5/2}^{3} (2z-5)\pi\, dz =$$

$$= (z^2\pi - 5z\pi)\Big|_{5/2}^{3} = 9\pi - \frac{25}{4}\pi - 15\pi + \frac{25}{2}\pi = -6\pi + \frac{25}{4}\pi$$

Ω_2: $\begin{cases} z \in [3,4] \\ 4 - x^2 - y^2 \ge z \end{cases}$ $t \in [0, 2\pi]$, jacobian ρ

$4 - \rho^2 \ge z \Rightarrow \rho^2 \le 4 - z \Rightarrow \rho \in [0, \sqrt{4-z}]$

$$\Rightarrow \iiint_{\Omega_2} dx\,dy\,dz = \int_3^4 dz \int_0^{2\pi} dt \int_0^{\sqrt{4-z}} \rho\, d\rho = \int_3^4 dz \int_0^{2\pi} \frac{\rho^2}{2}\Big|_0^{\sqrt{4-z}} dt =$$

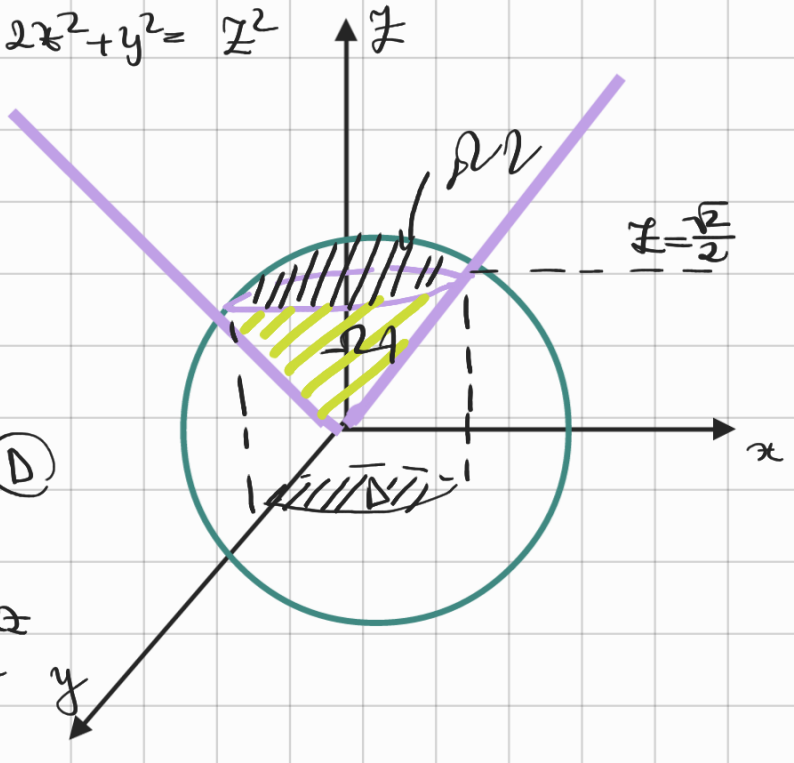
$$= \int_3^4 dz \int_0^{2\pi} \frac{(4-z)}{2}\, dt = \int_3^4 \frac{(4-z)}{2} \cdot t\Big|_0^{2\pi} dz = \int_3^4 (4-z)\,\pi\, dz = -\frac{(4-z)^2\pi}{2}\Big|_3^4$$

$$= \frac{\pi}{2}$$

$\Rightarrow$ Volumul lui Ω este $vol(\Omega_1) + vol(\Omega_2)$, adică

$$-6\pi + \frac{25}{4}\pi + \frac{\pi}{2} = \frac{-24\pi + 25\pi}{4} + \frac{\pi}{2} = \frac{\pi}{4} + \frac{\pi}{2} = \frac{3\pi}{4}$$

c) $2x^2 + y^2 + z^2 = 1$ și $2x^2 + y^2 = z^2$



$1 - 2x^2 - y^2 = 2x^2 + y^2 \Rightarrow$

$2x^2 + y^2 = \frac{1}{2} \Rightarrow z = \frac{\sqrt{2}}{2}$

$\Rightarrow \Omega$: $\begin{cases} z \in [\sqrt{2x^2+y^2}, \frac{\sqrt{2}}{2}] \\ 2x^2 + y^2 \le \frac{1}{2} \end{cases} \rightarrow (D)$

$\Rightarrow$ Volumul lui Ω este $\Omega_1/2$

$$\iiint_{\Omega} dx\,dy\,dz = \iint_D dx\,dy \int_{2x^2+y^2}^{1} 1\, dz$$

$$= \iint_D \left(\frac{\sqrt{2}}{2} - 2x^2 - y^2\right) dx\,dy$$

$D: \begin{cases} x = \frac{\sqrt{2}}{2}\rho\cos t \\ y = \rho \sin t \end{cases}$, $\rho \in [0, \frac{\sqrt{2}}{2}]$, $t \in [0, 2\pi]$ Jacobian $\frac{\sqrt{2}}{2}\rho \Rightarrow$

$t \in \left(\frac{\sqrt{2}}{2}\right)^{1-x^2-y^2}$

$$\iiint dx\,dy\,dz = \int_0^{\sqrt{2}/2} d\rho \int_0^{2\pi} \left(\frac{\sqrt{2}}{2} - \rho^2\right) \cdot \frac{\sqrt{2}}{2}\rho \, dt =$$

$$= \int_0^{\sqrt{2}/2} d\rho \int_0^{2\pi} \left(\frac{1}{2} - \rho^2 \cdot \frac{\sqrt{2}}{2}\right)\rho \, dt = \pi \int_0^{\sqrt{2}/2} (1 - \sqrt{2}\rho^2)\rho \, dt = \text{vol}(\Omega)$$

$$= \pi\left(\frac{\rho^2}{2} - \frac{\sqrt{2}\rho^4}{4}\right)\Big|_0^{\sqrt{2}/2} = \pi\left(\frac{1}{4} - \frac{\sqrt{2}}{16}\right) = \frac{\pi}{16}(4 - \sqrt{2})$$

(16) Aria suprafeței $x^2 + y^2 + z^2 = a^2$ situată în interiorul cilindrului $x^2 + y^2 = ax$, $a > 0$

$z^2 = a^2 - x^2 - y^2 \Rightarrow z = \sqrt{a^2 - x^2 - y^2} = f(x,y) \Rightarrow$

Aria suprafeței S este $A = \iint_D \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}\, dx\,dy$

unde $D: \{x^2 + y^2 \le ax \mid x, y \in \mathbb{R}\}$

$$\frac{\partial f}{\partial x} = \frac{-2x}{2\sqrt{a^2 - x^2 - y^2}} \Rightarrow \left(\frac{\partial f}{\partial x}\right)^2 = \frac{x^2}{a^2 - x^2 - y^2}$$

$$\left(\frac{\partial f}{\partial y}\right)^2 = \frac{y^2}{a^2 - x^2 - y^2} \Rightarrow$$

$$A = \iint_D \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}\, dx\,dy = \iint_D \sqrt{1 + \frac{y^2}{a^2 - x^2 - y^2} + \frac{x^2}{a^2 - x^2 - y^2}}\, dx\,dy$$

$$= \iint_D \frac{a}{\sqrt{a^2 - x^2 - y^2}}\, dx\,dy$$

Parametrizare D: $\begin{cases} x = \rho\cos t \\ y = \rho \sin t \end{cases}$ $\rho^2 \le a\rho\cos t \Rightarrow$ $\rho \ne 0$

$\rho \le a\cos t \Rightarrow$

$$\left.\begin{array}{l} t\in[0,\frac{\pi}{2}] \\ \rho\in[0,a\cos t] \end{array}\right\} \Rightarrow \quad \text{Jacobian } \rho$$

$$\iint_D \frac{a}{\sqrt{a^2-x^2-y^2}}\,dxdy = \int_0^{\pi/2} dt \int_0^{a\cos t} \frac{a}{\sqrt{a^2-\rho^2}}\cdot\rho\,d\rho$$

$$= \int_0^{\pi/2} -a\sqrt{a^2-\rho^2}\Big|_0^{\cos t}\,dt = \int_0^{\pi/2} -a\sqrt{a^2-a^2\cos^2 t}+a^2\,dt$$

$$= \int_0^{\pi/2} -a^2\sqrt{1-\cos^2 t}+a^2\,dt = a^2\int_0^{\pi/2}(1-\sqrt{\sin^2 t})\,dt =$$

$$a^2\int_0^{\pi/2}(1-\sin t)\,dt = a^2(t+\cos t)\Big|_0^{\pi/2} = a^2\left(\frac{\pi}{2}+0-0-1\right)$$

$$= a^2\left(\frac{\pi}{2}-1\right) \Rightarrow \text{cum cilindrul intersectează}$$

sfera în două suprafețe simetrice $\Rightarrow$ aria este $a^2(\pi-2)$.

(H.) Aria suprafeței $z=\sqrt{x^2+y^2}$, situată în interiorul cilindrului $x^2+y^2=2x$

$$z = f(x,y) = \sqrt{x^2+y^2}$$

$$\frac{\partial f}{\partial x} = \frac{x}{\sqrt{x^2+y^2}} \Rightarrow \left(\frac{\partial f}{\partial x}\right)^2 = \frac{x^2}{x^2+y^2} \quad \left(\frac{\partial f}{\partial y}\right)^2 = \frac{y^2}{x^2+y^2}$$

$\Rightarrow$ aria suprafeței este

$$\int_\Sigma d\sigma = \iint_D \sqrt{1+\frac{x^2}{x^2+y^2}+\frac{y^2}{x^2+y^2}}\,dxdy, \text{ unde}$$

$D: \; x^2+y^2\le 2x$ $\quad$ Jacobian ρ $\quad \begin{cases} x=\rho\cos t \\ y=\rho\sin t \end{cases}$ $\quad \rho^2\le 2\rho\cos t \Rightarrow$ $\rho\in[0,2\cos t]$, $t\in[0,\frac{\pi}{2}]$

$$\int_\Sigma d\sigma = \int_0^{\pi/2} dt \int_0^{2\cos t} \sqrt{2}\rho\, d\rho = \int_0^{\pi/2} \frac{\rho^2\sqrt{2}}{2}\Big|_0^{2\cos t} dt =$$

$$= \int_0^{\pi/2} \frac{4\cos^2 t\,\sqrt{2}}{2}\, dt = \sqrt{2}\int_0^{\pi/2} 2\cos^2 t\, dt = \sqrt{2}\int_0^{\pi/2}(1+\cos 2t)\,dt$$

$$= \sqrt{2}\left(t + \frac{1}{2}\sin 2t\right)\Big|_0^{\pi/2} = \sqrt{2}\cdot\frac{\pi}{2}$$

Cum cilindrul intersectează conul în 2 suprafețe identice $\Rightarrow$ aria suprafeței este $\pi\sqrt{2}$

Tipuri de exerciții examen arii

20. Să se calculeze aria $\mathcal{A}$ a suprafeţei Σ în următoarele cazuri:

a. $\Sigma: 2z = 4 - x^2 - y^2$, $z \in [0,1]$.

b. Σ este submulţimea de pe sfera $x^2 + y^2 + z^2 = 1$, situată în interiorul conului $x^2 + y^2 = z^2$.

c. Σ este submulţimea de pe sfera $x^2 + y^2 + z^2 = R^2$, situată în interiorul cilindrului $x^2 + y^2 - Ry = 0$.

d. Σ este submulţimea de pe paraboloidul $z = x^2 + y^2$ situată în interiorul cilindrului $x^2 + y^2 = 2y$.

e. Σ este torul.

Soluţie

Aria suprafeţei Σ este $\mathcal{A} = \int_\Sigma d\sigma$.

a. $\mathcal{A} = \int\int_D \sqrt{1 + x^2 + y^2}\,dxdy$, $D = \{(x,y)\,;\, 2 \le x^2 + y^2 \le 4\}$.

b. $\mathcal{A} = 2\int\int_D \frac{1}{\sqrt{1 - x^2 - y^2}}dxdy$, $D = \{(x,y) \in R^2 \mid x^2 + y^2 \le \frac{1}{2}\}$.

c. $\mathcal{A} = 2\int\int_D \frac{R}{\sqrt{R^2 - x^2 - y^2}}dxdy$, $D = \{(x,y)\,;\, x^2 + y^2 \le Ry\}$.

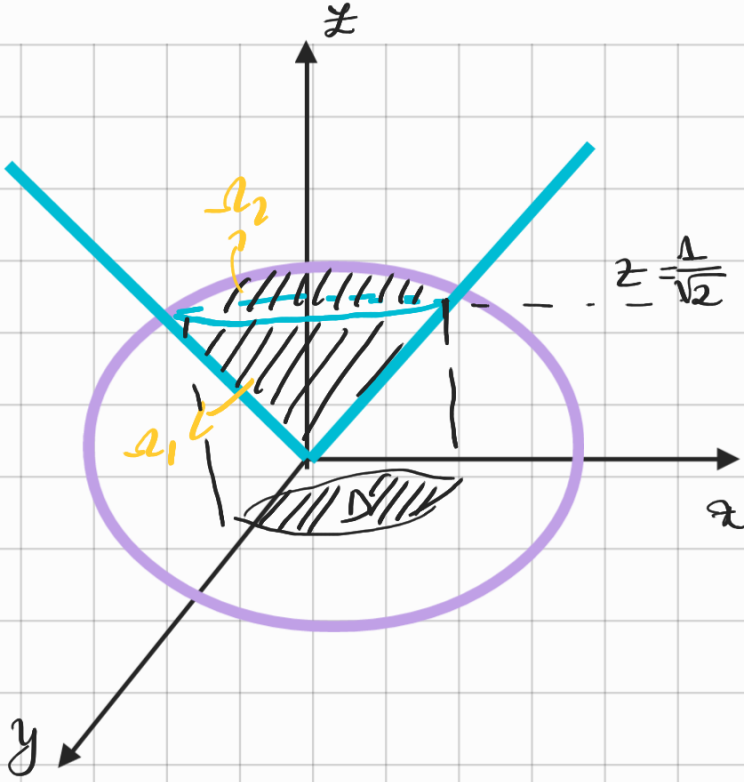
d. $\mathcal{A} = \int\int_D \sqrt{1 + 4x^2 + 4y^2}\,dxdy$, $D = \{(x,y)\,;\, x^2 + y^2 \le 2y\}$.

e. $\mathcal{A} = \int\int (a + b\cos u)dudv$, $D = [0, 2\pi] \times [0, 2\pi]$.

13. c) $2x^2+y^2+z^2=1$ și $2x^2+y^2=z^2$

↳ elipsoid ↳ con

Volumul domeniului dat Ω este $\text{vol}(\Omega_1)+\text{vol}(\Omega_2)$



$$\Omega_1:\begin{cases} z=z \\ x=\frac{1}{\sqrt{2}}\rho\cos t \\ y=\rho\sin t \end{cases}$$

$$\begin{cases} z\in[\sqrt{2x^2+y^2},\ \frac{1}{\sqrt{2}}] \\ \rho\in[0,\frac{1}{\sqrt{2}}] \\ t\in[0,2\pi] \end{cases} \text{ jacobian } \frac{1}{\sqrt{2}}\rho$$

$$\text{vol}(\Omega_1)=\iiint_{\Omega_1} dx\,dy\,dz=$$

$$=\iint_{\Delta}\left(\frac{1}{\sqrt{2}}-\sqrt{2x^2+y^2}\right)dx\,dy=$$

$$=\int_0^{2\pi}dt\int_0^{1/\sqrt{2}}\left(\frac{1}{\sqrt{2}}-\rho\right)\rho\frac{1}{\sqrt{2}}d\rho=$$

$$=\frac{1}{\sqrt{2}}\int_0^{2\pi}\left(\frac{\rho^2}{2\sqrt{2}}-\frac{\rho^3}{3}\right)\Big|_0^{1/\sqrt{2}}dt=$$

$$=\frac{1}{\sqrt{2}}2\pi\left(\frac{1}{4\sqrt{2}}-\frac{1}{6\sqrt{2}}\right)=\frac{1}{\sqrt{2}}\frac{\pi}{\sqrt{2}}\left(\frac{1}{2}-\frac{1}{3}\right)=\frac{1}{\sqrt{2}}\frac{\pi}{\sqrt{2}}\cdot\frac{3-2}{6}=\frac{\pi}{\sqrt{26}}=$$

$$=\frac{1}{\sqrt{2}}\cdot\frac{\pi-\sqrt{2}}{12}=\frac{\pi}{12}$$

Volumul este $\frac{\sqrt{2}\pi}{3}-\frac{\pi}{3}=\frac{\pi}{3}(\sqrt{2}-1)$

$$\Omega_2:\begin{cases} z=z \\ x=\frac{1}{\sqrt{2}}\rho\cos t \\ y=\rho\sin t \end{cases}$$

$$\begin{cases} z\in[\frac{1}{\sqrt{2}},\sqrt{1-2x^2-y^2}] \\ \rho\in[0,\frac{1}{\sqrt{2}}] \\ t\in[0,2\pi] \end{cases} \text{ jacobian } \frac{1}{\sqrt{2}}\rho$$

$$\text{vol}(\Omega_2)=\iiint_{\Omega_2} dx\,dy\,dz=\iint_{\Delta}\left(\sqrt{1-2x^2-y^2}-\frac{1}{\sqrt{2}}\right)dx\,dy=$$

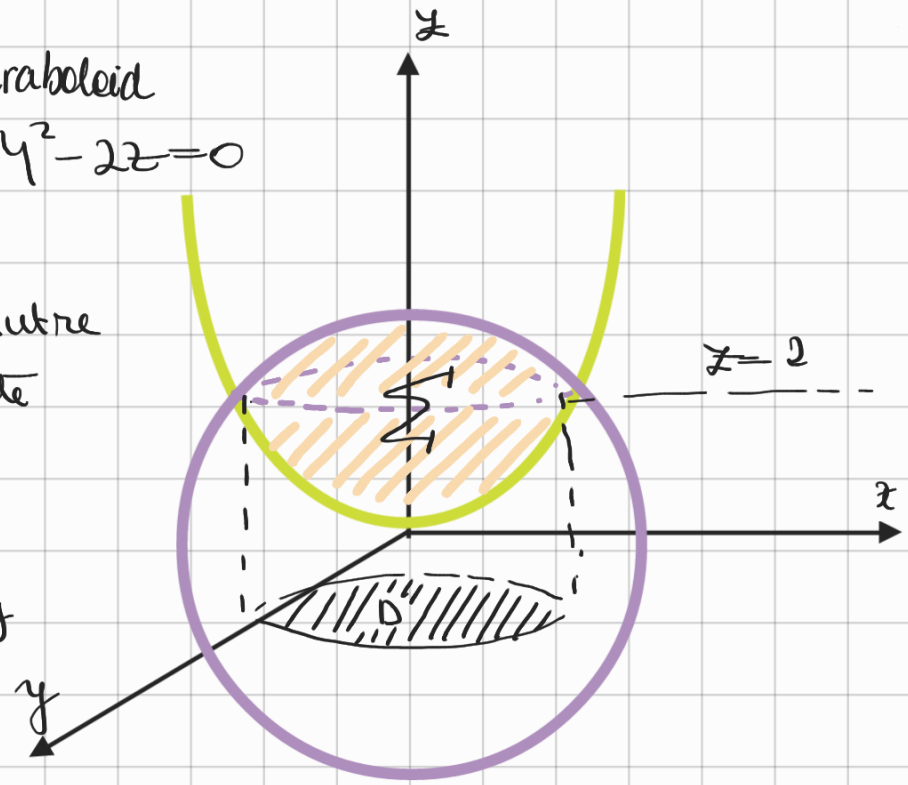
$$=\int_0^{2\pi}dt\int_0^{1/\sqrt{2}}\left(\sqrt{1-\rho^2}-\frac{1}{\sqrt{2}}\right)\cdot\frac{1}{\sqrt{2}}\rho\,d\rho=\frac{1}{\sqrt{2}}\int_0^{2\pi}dt\int_0^{1/\sqrt{2}}\left(\rho\sqrt{1-\rho^2}-\frac{1}{\sqrt{2}}\rho\right)d\rho$$

$$=\frac{1}{\sqrt{2}}\int_0^{2\pi}\left(1-\rho^2\right)^{3/2}\cdot\frac{2}{3}\cdot\frac{-1}{2}-\frac{1}{2\sqrt{2}}\rho^2\Big|_0^{1/\sqrt{2}}dt=\frac{2\pi}{\sqrt{2}}\cdot\left(\left(\frac{1}{2}\right)^{\frac{3}{2}}\cdot\frac{-1}{3}+\frac{1}{3}\right)-$$

$$-\frac{1}{2\sqrt{2}\cdot 2}\Big|=\frac{2\pi}{\sqrt{2}}\cdot\frac{1}{2\sqrt{2}}\cdot\frac{-1}{3}+\frac{1}{3}\cdot\frac{2\pi}{\sqrt{2}}-\frac{\pi}{4\sqrt{2}}=-\frac{\pi}{6}+\frac{\pi}{\sqrt{2}}=\frac{-2\pi-3\pi}{12}+\frac{\sqrt{2}\pi}{3}=\frac{-5\pi}{12}+\frac{\sqrt{2}\pi}{3}$$

13

a) $x^2+y^2+z^2=8$ (sferă), $z^2+y^2-2z=0$ (paraboloid)



Curba de intersecție dintre paraboloid și sferă este cercul de ecuație

$z=2,\ x^2+y^2=4 \rightarrow (D)$

Proiecția pe planul xOy a lui Σ_1 este

$D=\{(x,y) \mid x^2+y^2\leq 4\}$

$$\Rightarrow \text{vol}(\Omega)=\iiint_{\Omega} dxdydz=\iint_{D} dxdy \int_{\frac{x^2+y^2}{2}}^{\sqrt{8-x^2-y^2}} dz =$$

$$=\iint_{D}\left(\sqrt{8-x^2-y^2}-\frac{x^2+y^2}{2}\right) dxdy$$

Jacobian ρ

$$D:\begin{cases} x=\rho\cos t & \rho\in[0,2] \\ y=\rho\sin t & t\in[0,2\pi] \end{cases}\Rightarrow$$

$$\text{vol}(\Omega)=\int_0^{2\pi} dt \int_0^2 \left(\sqrt{8-\rho^2}-\frac{\rho^2}{2}\right)\cdot\rho\, d\rho = 2\pi\left(\frac{2^{\frac{9}{2}}-14}{3}\right)=$$

$$=2\pi\left(\frac{16\sqrt{2}-14}{3}\right)=\frac{4\pi}{3}(8\sqrt{2}-7)$$